Your grade: 100%

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Next item →

1. What is the range of values for probability? 1 / 1 point

- [ ] -1 to 1
- [ ] 0 to 10
- [ ] 0 to 100
- [x] 0 to 1

Nice work

Correct. Probability is a number between 0 and 1.

2. What is the probability of an impossible event? 1 / 1 point

- [ ] 0.75
- [ ] 1
- [x] 0
- [ ] 0.5

Nice work

Correct. An impossible event has a probability of 0.

3. If a coin is tossed, what is the probability of it landing heads up? 1 / 1 point

- [ ] 1
- [ ] 0.25
- [x] 0.5
- [ ] 0

Nice work

Correct. Assuming the coin is fair, the probability is 0.5 or 50 per cent.

4. Which of the following statements is true about mutually exclusive events? 1 / 1 point

- [ ] They are dependent on each other.
- [x] They cannot happen together.
- [ ] They can happen together.
- [ ] Their combined probability is always zero.

Nice work

Correct. By definition, mutually exclusive events cannot occur simultaneously.

5. What is the probability of rolling a 3 on a fair six-sided die? 1 / 1 point

- [x] 1/6
- [ ] 1/12
- [ ] 1/2
- [ ] 1/3

Nice work

Correct. There is one successful outcome out of six possible outcomes.

6. What does it mean when two events are independent? 1 / 1 point

- [ ] The events cannot happen at the same time.
- [ ] The occurrence of one event affects the occurrence of the other.
- [x] The occurrence of one event does not affect the occurrence of the other.
- [ ] The events are always equally likely.

Nice work

Correct. Independent events have no influence on each other.

7. If you roll two dice, what is the probability of both showing sixes? 1 / 1 point

- [x] 1/36
- [ ] 1/6
- [ ] 1/18
- [ ] 1/12

Nice work

Correct. The probability is 1/6 * 1/6 = 1/36.

8. What is the probability of rolling a number greater than four on a fair six-sided die? 1 / 1 point

- [ ] 1/2
- [ ] 1/6
- [x] 1/3
- [ ] 2/6

Nice work

Correct. There are two successful outcomes (5 and 6) out of six possible outcomes.

9. What is the probability of rolling an odd number on a fair six-sided die? 1 / 1 point

- [ ] 1/3
- [ ] 2/3
- [x] 1/2
- [ ] 1/6

Nice work

Correct. Half of the numbers on a six-sided die are odd.

10. What is the probability of rolling a sum of seven with two fair six-sided dice? 1 / 1 point

- [ ] 1/18
- [x] 1/6
- [ ] 1/12
- [ ] 1/36

Nice work

Correct. There are six successful outcomes out of 36 possible outcomes.

11. Calculate the probability of rolling a sum of nine with two fair six-sided dice. 1 / 1 point

1/9

Nice work

The probability of rolling a sum of nine with two fair six-sided dice is $\frac{1}{9}$.

1. Identify the possible outcomes for two dice: $(1,1),(1,2),\ldots,(6,6)$, which total 36 possible outcomes.

2. List all the pairs that sum to nine: $(3,6),(4,5),(5,4),(6,3)$, giving four successful outcomes.

3. Calculate the probability: $\frac{\text{4 successful outcomes}}{\text{36 possible outcomes}} = \frac{1}{9}$.

12. What is the probability of rolling at least one six when rolling two fair six-sided dice? 1 / 1 point

11/36

Nice work

The probability of rolling at least one six is $\frac{11}{36}$.

1. Calculate the probability of not rolling a six on a single die: $\frac{5}{6}$.

2. Calculate the probability of not rolling a six on either die: $\left(\frac{5}{6}\right)^2 = \frac{25}{36}$.

3. Subtract from one to find the probability of rolling at least one six: $1 - \frac{25}{36} = \frac{11}{36}$.

13. If a fair coin is tossed three times, what is the probability of getting exactly two heads? 1 / 1 point

3/8

Nice work

The probability of getting exactly two heads in three tosses is $\frac{3}{8}$.

1. List the possible outcomes: $HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$, which total eight possible outcomes.

2. Identify the successful outcomes: HHT, HTH, THH, giving three successful outcomes.

3. Calculate the probability: $\frac{\text{3 successful outcomes}}{\text{8 possible outcomes}} = \frac{3}{8}$.

14. Calculate the probability of rolling an even number on a fair six-sided die, then flipping a head on a fair coin. 1 / 1 point

1/4

Nice work

The probability of rolling an even number and flipping a head is $\frac{1}{4}$.

1. Probability of rolling an even number (2, 4 or 6): $\frac{3}{6} = \frac{1}{2}$.

2. Probability of flipping a head: $\frac{1}{2}$.

3. Multiply the probabilities: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

15. What is the probability of rolling a two or a four on a fair six-sided die? 1 / 1 point

1/3

Nice work

The probability of rolling a two or a four is $\frac{1}{3}$.

1. Probability of rolling a two: $\frac{1}{6}$.

2. Probability of rolling a four: $\frac{1}{6}$.

3. Add the probabilities (since they are mutually exclusive events): $\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$.